



Logophoricity and the processing of Chinese reflexives

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Abstract This study investigates the linguistic properties of two reflexive pronouns in Chinese, *ziji* ('self') and *ta-ziji* ('s/he-self'), and how the exempt use of these reflexives relates to the notion of perspective-taking or logophoricity. Using offline forced choice judgment and online self-paced reading experiments, we find that these two reflexives are more similar than typically assumed in the theoretical literature. Both reflexives are sensitive to the discourse topical status and the logophoric role (source/perceiver) of their long-distance antecedents. Relating discourse topicality and the Logophoric Hierarchy to perspective-taking, we posit that perspective-taking is a unifying factor that drives the exempt use of these two reflexives. However, these reflexives differ slightly in that *ta-ziji* requires a more prominent perspective center than *ziji* does. In addition, the current study shows that during incremental processing, different types of discourse information can be accessed at different processing stages, which contributes to our understanding of the accessibility of linguistic cues during real-time anaphora resolution.

Keywords Chinese · Reflexive pronoun · Exempt reflexive · Logophoricity · Sentence processing

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1 Introduction

Reflexive pronouns (e.g., *herself*) depend on other referential expressions—antecedents—for their meaning. However, the search for an antecedent is constrained by various linguistic factors. For example, Principle A of the Binding Theory (Chomsky 1981, 1986) dictates that reflexives can only be bound by a structurally prominent (i.e., c-commanding) antecedent in the local domain. But a large body of work suggests that syntactic prominence is only one of the various factors that influence the construal of reflexives (e.g., Cantrall 1974; Zribi-Hertz 1989; Pollard and Sag 1992; Reinhart and Reuland 1993). In this paper, we focus on the role of discourse prominence in reflexive resolution; in particular, how *perspective-taking* guides reflexive resolution in Chinese.

The relevance of perspective for reflexive resolution is illustrated by the literature on *logophoricity*. Logophoricity refers to the property of some pronouns, called *logophoric pronouns*, which are anchored to individuals whose thoughts, feelings, and general states of consciousness are reported (Hagège 1974; Clements 1975). Some West African languages have morphologically distinct logophoric pronouns, but it has also been suggested that long-distance reflexives in languages such as Icelandic (e.g., Maling 1984; Sigurðsson 1990), Japanese (e.g., Kuroda 1965; Kuno 1972; Oshima 2007; Nishigauchi 2014), and Chinese (e.g., Huang and Liu 2001; Pollard and Xue 2001) are perspective-sensitive. Due to this connection, long-distance reflexives referring to perspective centers are widely called *logophoric reflexives* (e.g., Reinhart and Reuland 1993; Huang and Liu 2001) or *exempt reflexives* (e.g., Pollard and Sag 1992; Charnavel and Sportiche 2016). In this paper, we use the general terms *exempt reflexive* or *perspective-sensitive reflexive* when discussing Chinese long-distance reflexives.

With respect to logophoricity and long-distance construal of Chinese reflexives, the common assumption is that the simplex reflexive *ziji* ('self') can have an exempt use—sensitive to logophoricity—but that the complex reflexive *ta-ziji* ('s/he-self') is a local anaphor constrained by Principle A (e.g., Huang 1983; Battistella and Xu 1990; Huang and Tang 1991; Huang and Liu 2001). (We follow Pollard and Xue (1998) by assuming that there is only one *ziji* which can have two uses.) This difference between *ziji* and *ta-ziji* is illustrated by ex.(1) where *ziji* can have a local or non-local reading, while *ta-ziji* has a strong preference for the local reading.

- (1) Xiaoming₁ tingshuo Wang jiaoshou₂ yuedu-le *ziji*_{1/2}/*ta-ziji*_{*1/2} de lunwen.
 Xiaoming hear Professor Wang read-ASP self/he-self DE paper
 Xiaoming heard that Prof. Wang had read self's paper.

The non-local reading of *ziji* is licensed when the embedded clause is viewed from the perspective of the matrix subject *Xiaoming* (e.g., Huang and Liu 2001; Pan 2001), as is expected if we are dealing with a perspective-sensitive reflexive. The non-local reading of *ta-ziji*, in contrast, is blocked by the local referent 'Prof. Wang.' However, Pan (1998) noticed that if the local subject is inanimate, non-local binding is allowed, shown in (2):

- (2) Xiaoming zhidao na-ben shu huai-le ta-ziji de mingyu.
 Xiaoming know that-CL book damage-ASP he-self DE reputation.
 ‘Xiaoming knows that the book damaged self’s reputation.’

This animacy blocking effect exhibited by *ta-ziji* is captured by the Anaphor Condition (Pan 1998, 794) which states that long-distance binding is only licensed when the non-local referent is semantically more prominent than the local referent, where semantic prominence is determined by an Animacy Hierarchy (Chou 1992). It is worth mentioning that Pan assumes that *ta-ziji* is a ‘plain’, non-logophoric anaphor (Pan 1998, 811–813), except that its binding properties cannot be fully predicted by Principle A. Nevertheless, given a large body of literature on logophoricity and complex reflexives in English (e.g., Zribi-Hertz 1989; Pollard and Sag 1992), French (e.g., Charnavel 2019, 2020), and Korean (e.g., Kim and Yoon 2009), it is reasonable to speculate that *ta-ziji* may be sensitive to some kind of discourse (e.g., perspective-related) prominence.

In fact, ex.(2) could be taken as evidence that perspective is relevant for the long-distance construal of *ta-ziji*. According to the Humanness Empathy Hierarchy (Kuno and Kaburaki 1977; Kuno 1987), people tend to empathize with humans rather than inanimate entities. The term *empathy* involves “speaker’s identification with a person that participates in an event or state” (Kuno 1987, 206) and, for the purposes of this paper, can be construed synonymously with *perspective-taking*. Thus, one potential explanation for the acceptability of ex.(2) is that comprehenders view the sentence from the perspective of *Xiaoming*—i.e., he is the perspective center and can thus be an antecedent for *ta-ziji*. Along similar lines, the unavailability of non-local reading for *ta-ziji* in ex.(1) could be attributed to *Xiaoming* not being sufficiently prominent relative to ‘Prof. Wang’ to function as a perspective center. This line of reasoning brings up the possibility that *ta-ziji* can also be logophoric, contrary to what is typically assumed. The logophoric properties of *ta-ziji* and the question of how *ta-ziji* differs from *ziji* are two key linguistic questions that this paper aims to shed light on.

In this study, we examine the logophoric properties of *ziji* and *ta-ziji* by probing Chinese speakers’ offline antecedent choices (Exp. 1a, 2a) and their real-time processing patterns (Exp. 1b, 2b). While offline experiments provide valuable insights into participants’ final judgments, online (i.e., real-time) experiments allow us to have a dynamic view of reflexive resolution as sentences are processed incrementally, thus providing more information on *when* discourse information is utilized during reflexive resolution. The question of when syntactic and non-syntactic (e.g., semantic and discourse) information are used during online reflexive resolution is hotly debated among psycholinguists (e.g., Sturt 2003; Runner et al. 2006; Kaiser et al. 2009; Cunnings and Sturt 2014; Patil et al. 2016; Sloggett 2017). Thus, in addition to shedding light on linguistic questions, we aim for this study to have implications for the psycholinguistic aspects of reflexive resolution as well.

The rest of the paper is organized as follows. Section 2 reviews existing work on logophoricity and Chinese reflexives and shows how we can use discourse topicality

and the Logophoric Hierarchy (Culy 1994) to probe whether and how perspective impacts the processing of *ziji* and *ta-ziji*. Section 3 reviews psycholinguistic debates on discourse prominence and online reflexive resolution. Section 4 is an overview of our research goals. We then report two sets of offline and online experiments in Sects. 5 and 6, and discuss the broader implications of this study in Sect. 7. Section 8 concludes the paper.

2 Logophoricity and Chinese reflexives

In this section, we review linguistic evidence that long-distance *ziji* is perspective-sensitive and highlight the open question of whether *ta-ziji* can also be perspective-sensitive. We then introduce the concepts of discourse topicality (Section 2.1) and the Logophoric Hierarchy (Section 2.2) before testing in a series of psycholinguistic experiments whether, how, and when *ziji* and *ta-ziji* are sensitive to perspectival information during incremental sentence comprehension.

Several properties of long-distance *ziji* suggest that it is anchored to a perspective center. First, the content of an embedded clause containing *ziji* reflects the first-personal mental perspective of the non-local antecedent. In other words, *ziji* must have a *de se* reading in attitude contexts (e.g., Pan 1997; Huang and Liu 2001). Second, long-distance *ziji* is subject to person feature blocking: a 1st-person pronoun *wo* ‘I’ reflecting the actual speaker’s perspective blocks reference to a long-distance sentence-internal 3rd-person perspective holder (e.g., Huang et al. 1984; Xue et al. 1994; Pan 1997; Pollard and Xue 1998; Huang and Liu 2001; but see e.g., Cole and Sung 1994; Cole and Wang 1996; Giblin 2016 for different views). Thus, if ‘Prof. Wang’ in ex.(1) is replaced by *wo* ‘I’, non-local binding of *ziji* by *Xiaoming* becomes unavailable because speakers prioritize their own perspectives than others’ (Kuno and Kaburaki 1977; Kuno 1987; Huang and Liu 2001). Finally, multiple instances of long-distance of *ziji* in the same perspectival domain must refer to the same perspective center due to perspective uniformity (e.g., Pan 1997; Charnavel 2019, 2020). In short, there is convincing evidence that long-distance *ziji* is perspective-sensitive.

Unlike *ziji*, long-distance *ta-ziji* is often unacceptable, especially when the local subject is animate/human. However, a recent experimental study by Liu (2020) presents compelling preliminary evidence that *ta-ziji* may be perspective-sensitive. In her study, Liu compared non-local binding of *ta-ziji* by an attitude holder and by a non-attitude-holder, respectively. We repeat her examples (Liu 2020: ex.(6)) in ex.(3a–b):

- (3) a. Ju Lisi shuo, zhe-jian shi shanghai-le ta-ziji.
 According to Lisi say this-CL matter hurt-ASP he-self
 ‘According to Lisi, this matter hurt self.’
 b. Shuodao Lisi, zhe-jian shi shanghai-le ta-ziji.
 Speaking of Lisi this-CL matter hurt-ASP he-self
 ‘Speaking of Lisi, this matter hurt self.’

In a binary acceptability judgment experiment, Liu found that Chinese speakers prefer (3a) over (3b), which she attributes to the fact that *Lisi* is a perspective holder in (3a) but not in (3b). In addition, long-distance binding of *ta-ziji* inside a perspective-expressing *because*-clause (Kuroda 1965; Sells 1987; Huang and Liu 2001; Nishigauchi 2014) is largely acceptable (see ex.(4)), although the acceptability of long-distance *ta-ziji* was found to be lower compared to long-distance *ziji* in this type of sentence (Liu 2020).

- (4) Lisi₁ likai-le gongsi, yinwei nū jingli piping-le ta-ziji₁.
 Lisi leave-ASP company because female manager scold-ASP she-self
 ‘Lisi left the company because the female manager scolded self.’

However, these findings are subject to alternative explanations. For example, the lower acceptability of (3b) relative to (3a) may be related to the subject orientation constraint since *Lisi* in (3b) occupies the object position. Furthermore, the acceptability of non-local binding in ex.(4) could be due to the use of other-directed verbs like ‘scold’ (more on this in Section 2.1) which semantically bias the non-local subject (also see Yu 1992, 288). Previous studies show that although native speakers are heavily influenced by verb semantics in their sentence-final judgments, their online reactions upon encountering the reflexive may be different (e.g., Lu 2011; Lyu and Kaiser 2023). Therefore, Liu’s study results linking logophoricity to the exempt reading of *ta-ziji* remain inconclusive. In the present study, we approach this topic from two angles by testing how discourse topicality and the Logophoric Hierarchy—two concepts closely linked to the notion of perspective—impact the interpretation of *ta-ziji* in addition to *ziji*.

2.1 Discourse topicality and perspective-sensitive reflexives

We start this subsection by introducing Kuno’s (1987) Topic Empathy Hierarchy and how it mediates perspective-taking in reflexive resolution. Kuno (1987, 210) noticed that, in ex.(5) with a topical sentence introducing Mary into the discourse, only continuation (5a), but not (5b), is felicitous. This is because it is “difficult for the speaker to empathize with someone he has just introduced into the discourse to the exclusion of someone he has been talking about in the preceding discourse” (Kuno and Kaburaki 1977, 642). This captures the essence of the Topic Empathy Hierarchy, defined in ex.(6).

- (5) Mary had quite an experience at the party she went to last night.
 a. She met an eight-foot-tall basketball player.
 b. *An eight-foot-tall basketball player met her.
- (6) Topic Empathy Hierarchy (Kuno 1987, 210)
 Given an event or state involving A and B such that A is referential with the topic of the present discourse and B is not, it is easier for the speaker to empathize with A than B.

The Topic Empathy Hierarchy thus points to a clear link between discourse topicality and perspective-sensitivity. Indeed, prior studies show that discourse topics tend to be construed as perspective centers (e.g., Pollard and Sag 1992; Oshima 2007; Hinterwimmer 2019; von Heusinger and Schumacher 2019) and that this can have grammatical consequences. For instance, in Japanese, the inclusion of an extra perspective center through an empathic morpheme like *kureru* ('give'), in addition to the existing discourse topic (also a perspective center), makes the sentence unacceptable (Oshima 2007, 22). Thus, building on Kuno's Topic Empathy Hierarchy, we hypothesize that *a perspective-sensitive reflexive should show a preference to refer to the discourse topic (i.e., the perspective center) over a non-topical referent*. In fact, evidence concerning *ziji* (e.g., Lyu and Kaiser 2021, 2023) suggests that this linking hypothesis is on the right track. We illustrate this with an example from Lyu and Kaiser (2021), shown in ex.(7):

- (7) Xiaoming shi banji li de youxiu xuesheng. Ke shang, ta
 Xiaoming be class in DE excellent student Lecture on he
 tingshuo Wang jiaoshou
 hear Prof. Wang
 {fabiao_{self-directed}-le/pigai_{other-directed}-le} ziji de xueshu lunwen.
 publish_{self-directed}-ASP/grade_{self-directed}-ASP self DE academic paper
 'Xiaoming is an excellent student in the class. During the lecture,
 he heard that Prof. Wang had
 {published_{self-directed}/graded_{other-directed}} self's academic paper.'

In a self-paced reading experiment where the non-local referent (e.g., 'Xiaoming') is presented as a discourse topic, Lyu and Kaiser (2021) found that Chinese speakers exhibited reading-time slowdowns when the sentence contained a self-directed verb (e.g., 'publish') that biases the local referent (e.g., 'Prof. Wang'), compared to when the sentence contained an other-directed verb (e.g., 'grade') that biases the non-local referent. They assumed that reading-time slowdowns reflect a conflict between binding preferences and verb semantics. Thus, the reading slowdowns elicited by self-directed verbs suggest that Chinese speakers anticipate non-local binding of *ziji* (which clashes with the semantic bias of self-directed verbs) in the presence of a discourse topic (likely to be construed as a perspective center). This aligns with the idea that *ziji* is perspective-sensitive.

However, what remains unclear is the discourse-pragmatic property of *ta-ziji*. Lyu and Kaiser (2023) used the same topicality design and the verb bias diagnostic as in ex.(7) to examine the processing patterns of *ta-ziji*. Intriguingly, they observed that when the non-local referent is a discourse topic, *ta-ziji* shows no binding preference (no verb bias effect), but when the non-local referent is not a discourse topic, *ta-ziji* exhibits a local binding preference, with other-directed verbs causing reading slowdowns. Although this suggests that discourse topicality impacts the interpretation of *ta-ziji*, the absence of any binding preference for *ta-ziji* in sentences with topical subjects like ex.(7) leads Lyu and Kaiser (2023) to the tentative

conclusion that the topicality effect for *ta-ziji* is unrelated to perspective-taking. This argument seems plausible since Pollard and Xue (2001) postulate that various discourse factors—including topicality, contrastiveness, and logophoricity—can all play a role in the construal of reflexives.

However, we think that Lyu and Kaiser’s (2023) conclusion may be premature. One alternative explanation for the weaker topicality effect of *ta-ziji*, compared to *ziji*, could be that *ta-ziji* requires a more prominent perspective center. This speculation seems reasonable as previous studies show that pronominals with different morphological forms display different syntactic and pragmatic properties (e.g., Brown-Schmidt et al. 2005; Kaiser and Trueswell 2008; Kaiser et al. 2009). To further explore the relationship between logophoricity and long-distance reflexives in Chinese, in the present work we aim not only to replicate the two reflexives’ sensitivity to discourse topicality, but also follow previous experimental studies (e.g., Kaiser and Runner 2008; Kaiser et al. 2009; Han et al. 2015; Sloggett 2017) by directly testing how the Logophoric Hierarchy (Culy 1994) influences reflexive resolution in Chinese, which we turn to next.

2.2 The Logophoric Hierarchy and perspective-sensitive reflexives

Before we explain how we use Culy’s (1994) Logophoric Hierarchy to examine the logophoric properties of *ziji* and *ta-ziji*, we acknowledge Sells’ (1987) division of logophoric roles into *source* (i.e., intentional speaker), *self* (i.e., one whose mental state is reported), and *pivot* (i.e., deictic center), which interact with the binding of exempt reflexives (e.g., Huang and Liu 2001; Cole et al. 2001; Kim and Yoon 2009). However, for the purposes of our study, we use Culy’s Logophoric Hierarchy, shown in ex.(8), to make our findings comparable with prior work (e.g., Kaiser and Runner 2008; Kaiser et al. 2009; Sloggett 2017).

(8) The Logophoric Hierarchy (Culy 1994, 1065)

Speech > thought > knowledge > direct perception

According to the Logophoric Hierarchy in (8), subjects of speech predicates (*sources* of information) are most likely to antecede logophoric pronouns, while subjects of perception predicates (*perceivers* of information) are least likely to antecede logophoric pronouns. Culy’s Logophoric Hierarchy was originally based on typological investigations of ‘pure’ logophoric languages in which logophoric pronouns cannot have a reflexive use, instead of ‘mixed’ logophoric languages whose reflexives exhibit perspective-sensitive properties. However, Culy’s hierarchy turns out to interact with exempt reflexives in mixed logophoric languages as well.

For instance, Maling (1984, 232, ex.(37–38)) discusses examples from Icelandic where a source but not a perceiver can antecede the logophoric reflexive *sig*. This observation is echoed by crosslinguistic experimental studies on exempt reflexives (e.g., Kaiser and Runner 2008; Kaiser et al. 2009; Han et al. 2015; Sloggett 2017; Kaiser 2022), which show that, probabilistically, sources are more likely to antecede exempt reflexives than perceivers in both offline and online experiments. Thus,

unlike ‘pure’ logophoric languages in which the Logophoric Hierarchy is categorical and grammaticalized, ‘mixed’ logophoric languages show gradient sensitivity to this hierarchy.

In the present study, we follow previous experimental work by comparing the binding of Chinese reflexives by antecedents with source and perceiver roles.¹ We manipulate these two roles using the verbs ‘tell’ vs. ‘hear.’ This kind of verb manipulation has been successfully used in prior work on English, German and Dutch (e.g., Kaiser and Runner 2008; Kaiser et al. 2009; Sloggett 2017). See ex.(9) for an example:

- (9) Xiaoming shi banji li de youxiu xuesheng. Ke shang,
 Xiaoming be class in DE excellent student Lecture on
 ta {gaosu bieren/tingshuo}
 he tell others hear-say
 Wang jiaoshou {fabiao_{self-directed}/pigai_{other-directed}}-le ziji/ta-ziji de
 xueshu lunwen.
 Prof. Wang publish_{self-directed}-ASP/grade_{other-directed}-ASP self/he-self DE
 academic paper
 ‘Xiaoming is an excellent student in the class. During the lecture, he {told
 others/heard} that
 Prof. Wang had {published_{self-directed}/graded_{other-directed}} self’s academic paper.’

In ex.(9), the matrix subject (‘he’—*Xiaoming*) of the second sentence is a topic introduced in first sentence. Given the Topic Empathy Hierarchy, *Xiaoming* can be construed as a perspective center. Furthermore, based on the Logophoric Hierarchy, sources should be especially suitable to be perspective centers compared to perceivers. Thus, when determining the antecedent of the reflexive, Chinese speakers may be more likely to select a source than a perceiver. Additionally, when processing the reflexive, they may show a stronger preference for a source than a perceiver.

As discussed in Section 2.1, processing effects can be detected by manipulating the verb in the second clause—specifically, by comparing self-directed vs. other-directed verbs. If people prefer to interpret a reflexive as non-locally bound, then encountering this reflexive after a self-directed verb should elicit reading slowdowns. This is because a self-directed verb creates an expectation for a *locally*-bound construal, and if the reflexive form has preference for a *non-local* construal, this results in a conflict. If sources are better non-local antecedents than perceivers, we may find that self-directed verbs elicit slower reading times when the topical subject is a source than a perceiver. In essence, we use verb bias effects as a

¹ An anonymous reviewer suggests that the concepts of source and perceiver can be replaced with the concept of attitude holder. However, we think it is useful to introduce fine-grained distinctions into different subtypes of attitude holders given a large body of literature indicating that sources are more prototypical perspective centers compared to perceivers and other logophoric roles (e.g., Sells 1987; Huang and Liu 2001; Kaiser and Runner 2008; Kim and Yoon 2009; Kaiser et al. 2009; Han et al. 2015; Sloggett 2017; Kaiser 2022).

diagnostic for how strongly a reflexive prefers a local vs. non-local interpretation. As will become clear below, the verb bias manipulation also allows us to tap into effects of discourse topicality. The design illustrated by ex.(9) is therefore an improvement over Lyu and Kaiser's (2023) study which did not control the logophoric role (source/perceiver) of the topical referent and thus, as discussed in Section 2.1, does not yield conclusive evidence regarding logophoricity and *ta-ziji*.

3 Accessibility of discourse information during online reflexive resolution

So far, we have considered the relationship between logophoricity and Chinese reflexives. However, our study is also concerned with how different types of information are integrated and utilized during real-time reflexive resolution. We focus on two key questions: (i) Is syntactic information prioritized over discourse information? (ii) Do all types of discourse-pragmatic cues (e.g., discourse-topical status, logophoric roles) have similar impacts on the processing of *ziji* and *ta-ziji*? In this subsection, we review debates on these two central psycholinguistic questions.

Critical for the purposes of this study, there is convincing evidence that syntactic cues (e.g., syntactic locality) are 'defeasible' during real-time processing of reflexives. Previous online studies of reflexive resolution suggest that a topically prominent referent (Sturt 2003; Cunnings and Sturt 2014; Patil et al. 2016) or logophorically prominent referent (e.g., Kaiser et al. 2009; Han et al. 2015; Sloggett 2017) may be considered as a potential antecedent of the reflexive *himself/herself* during *some* stage of processing. But there are debates on *when* discourse-level cues relative to syntactic cues are accessed and integrated into mental representation during incremental processing.

The first approach—the *syntax-first account*—posits that syntactic information/syntactic cues are accessed earlier during processing than discourse-level cues such as discourse topicality and logophoric roles of the antecedent. Several studies find that when a topically prominent referent (*Jonathan/Jennifer* in ex.(10)), introduced in prior discourse, is outside the local domain, native English speakers do not consider this referent as the antecedent during the early stages of processing (e.g., Sturt 2003; Felser and Cunnings 2012).

- (10) Jonathan/Jennifer was pretty worried at the city hospital. He/she remembered that the surgeon had pricked himself with a used syringe needle. (Sturt 2003, 546)

In fact, similar findings involving picture-NP reflexives, known to be exemptable from Principle A (e.g., Jackendoff 1972; Cantrall 1974; Pollard and Sag 1992), also suggest that local referents are considered first despite the discourse prominence of the non-local referent, at least in English (Cunnings and Sturt 2014). However, existing evidence on online reflexive resolution in Chinese (e.g., Lyu and Kaiser 2021, 2023) suggests otherwise. (But it could be that the syntax-first account works better for certain languages due to language-specific properties—that

question is beyond the scope of this paper.) Thus, we follow prior work and hypothesize that, at least in Chinese, syntactic locality may not be prioritized in real-time reflexive resolution in discourse.

In contrast to the syntax-first account, other results are more in line with *multiple-constraints approaches* of reflexive resolution (e.g., Kaiser et al. 2009; Han et al. 2015; Patil et al. 2016; Sloggett 2017). Multiple-constraints approaches, as they have been proposed in the literature, have two key characteristics that are relevant for our work. First, they posit that *multiple kinds of information* can be immediately accessible during the process of reference resolution. For example, in a visual-world eye-tracking study using sentences like *Peter told/heard from Andrew about a picture of himself*, Kaiser et al. (2009) discovered that, upon hearing the reflexive, native English speakers looked more at pictures of source subjects than perceiver subjects and that this preference emerged early. This suggests that certain discourse information (e.g., logophoric roles) is accessible at the earliest moments of reflexive resolution. Further evidence from reading eye-tracking studies (Patil et al. 2016; Sloggett 2017) strengthens this view. The present study involving discourse prominence is especially appropriate for examining the timing of when different kinds of discourse constraints impact real-time processing, since discourse plays a pivotal role in reference resolution in Chinese (e.g., Li and Thompson 1981).

The second key characteristic supported by some studies that favor multiple-constraints approaches to reference resolution is the *form-specific* nature of the process. This is explicitly assumed in the *form-specific multiple-constraints hypothesis* (e.g., Kaiser and Trueswell 2008; Kaiser et al. 2009). According to this account, different referential forms can show different levels of sensitivity to different kinds of information. In our case, it could be that *ziji* and *ta-ziji* show differing sensitivities to two kinds of discourse-level information, e.g., discourse topicality and the Logophoric Hierarchy. In other words, one type of discourse-level information could have a bigger impact—in terms of the magnitude or the time-course of the effects—on reflexive resolution than the other type of discourse-level information.

It is worth emphasizing that while the form-specific multiple-constraints hypothesis freely allows for a situation where different anaphoric forms differ in their sensitivities to different kinds of syntactic, semantic, and discourse-level information, prior work has focused on comparing different classes of constraints (especially syntactic vs. semantic/pragmatic constraints) and how sensitive different forms are to these different classes of constraints. But many aspects of possible form-specific effects remain untested. In particular, to our knowledge, there is little experimental work systematically testing whether different anaphoric forms can differ in how sensitive they are to different *subtypes* of discourse-level information. We seek to address this gap.

Summing up, based on earlier work, in the current investigation of the Chinese reflexives *ziji* and *ta-ziji*, we consider two kinds of multiple-constraints approaches. On the one hand, reference resolution could be guided by multiple constraints from the earliest moments onwards, with both forms being equally sensitive to multiple constraints. In this situation, there are no form-specific differences; the forms pattern uniformly. In our case, it could be that *ziji* and *ta-ziji* are both equally sensitive to two kinds of discourse-level information, namely discourse topicality

and the Logophoric Hierarchy, without exhibiting any form-specific differences. We call this view the *uniform multiple-constraints approach*. On the other hand, as mentioned above, reference resolution can show form-specific sensitivity to different factors, with *ziji* and *ta-ziji* exhibiting differing sensitivities to discourse topicality and/or the Logophoric Hierarchy. Such an outcome would support a *form-specific multiple-constraints approach*.

In this study, the uniform multiple-constraints hypothesis and the form-specific multiple-constraints hypothesis are our primary foci in evaluating the influence of discourse topicality and logophoric roles on online reflexive resolution in Chinese. Findings pertaining to these hypotheses could enrich our understanding of real-time processing of reflexives.

4 Overview of the aims

This study has two main goals, a linguistic one and a psycholinguistic one, outlined below.

Our linguistic goal is to shed light on the relationship between Chinese exempt reflexives and the Logophoric Hierarchy and how the discourse-pragmatic properties of *ziji* and *ta-ziji* differ. Although prior studies on exempt reflexives may lead us to expect that sources are more likely to be taken as antecedents (perspective centers) (e.g., Kaiser and Runner 2008; Kaiser et al. 2009; Sloggett 2017), the extent to which this holds true for Chinese reflexives remains to be tested. Furthermore, whether the complex reflexive *ta-ziji* can be used as a logophoric reflexive is far from being clear, given the different views in the field (e.g., Huang and Tang 1991; Yu 1992; Pan 1998; Charnavel 2019, 85; Liu 2020).

The psycholinguistic goal of this study is to examine the relative accessibility of the two discourse cues in real-time reflexive resolution. Setting aside the syntax-first account which predicts local binding preferences for both reflexives, the uniform and form-specific multiple-constraints accounts differ in predictions. Assuming that discourse topicality and the Logophoric Hierarchy indeed modulate the linguistic behaviors of *ziji* and *ta-ziji*, the uniform multiple-constraints account predicts that discourse topicality and logophoric roles immediately guide online resolution of *ziji* and *ta-ziji* to an equal extent. By contrast, the form-specific approach has the flexibility of allowing for these two types of discourse information to impact the processing of *ziji* and *ta-ziji* differently in terms of the magnitude and/or time-course of the discourse-related effects.

In Section 5, we report two sets of experiments on the processing of *ziji* and *ta-ziji*, respectively. Each set of experiments comprises an offline forced choice task (Exp. 1a, 2a) and a self-paced reading task (Exp. 1b, 2b).

5 Experiment 1: logophoricity and the processing of *ziji*

Experiment 1 examines whether the exempt use of *ziji* interacts with the Logophoric Hierarchy. The offline forced choice Experiment 1a taps into speakers' overall

preferences after the sentence has been fully processed. Experiment 1b examines the time-course of accessibility of discourse topicality and logophoric roles in online processing of *ziji*.

5.1 Materials and design

These experiments build on but extend the experimental design of Lyu and Kaiser (2023). Experiment 1a (forced choice) and 1b (self-paced reading) use the same set of stimuli. Each target item is composed of two sentences, a context sentence introducing a topical referent into the discourse and a critical sentence about that topical referent. (Given the Topic Empathy Hierarchy, the topical referent can be viewed as a perspective center.) Thus, our targets are designed so that the non-local antecedent is always a discourse topic. We chose to make the non-local subject topical because prior work on Chinese reflexives suggests that, without a context that biases the non-local antecedent as a discourse topic (perspective center), native speakers tend to interpret *ziji* as a plain non-logophoric anaphor that is locally bound (e.g., Dillon et al. 2014, 2016; Jäger et al. 2015; Wang 2017a, b). (See more on discourse topicality below.) For the design of the critical sentences, LOGOPHORIC ROLE (source/perceiver) and VERB BIAS (self-directed/other-directed) were crossed to yield a 2×2 design. See ex.(11) for an example set of critical sentences.

(11) Context sentence:

Xiaoming shi banji li de youxiu xuesheng.
Xiaoming be class in DE excellent student
'Xiaoming is an excellent student in the class.'

Critical sentence:

a. Source/self-directed

Ke shang, ta gaosu bieren Wang jiaoshou fabiao-le
Lecture on he tell others Prof. Wang publish-ASP
ziji de xueshu lunwen.
self DE academic paper
'During the lecture, he told others that Prof. Wang had published self's academic paper.'

b. Source/other-directed

Ke shang, ta gaosu bieren Wang jiaoshou pigai-le
Lecture on he tell others Prof. Wang grade-ASP
ziji de xueshu lunwen.
self DE academic paper
'During the lecture, he told others that Prof. Wang had graded self's academic paper.'

c. Perceiver/self-directed

Ke shang, ta tingshuo Wang jiaoshou fabiao-le ziji
Lecture on he hear-say Prof. Wang publish-ASP self
de xueshu lunwen.
DE academic paper

‘During the lecture, he heard that Prof. Wang had published self’s academic paper.’

d. Perceiver/other-directed

Ke	shang,	ta	tingshuo	Wang jiaoshou	pigai-le	ziji
Lecture	on	he	hear-say	Prof. Wang	grade-ASP	self
de	xueshu	lunwen.				
DE	academic	paper				

‘During the lecture, he heard that Prof. Wang had graded self’s academic paper.’

As discussed in Section 2.2, perspective-sensitive reflexives may show gradient sensitivity to the Logophoric Hierarchy such that sources are more likely to be treated as perspective centers relative to perceivers. Thus, we might anticipate stronger non-local binding preferences for sources in (11a-b) than for perceivers in (11c-d). In the offline forced choice Experiment 1a, the strength of this preference is reflected in the proportion of non-local antecedent choices.

In the online Experiment 1b, we follow prior studies (e.g., Lu 2011; Wang 2017a, 2017b; Lyu and Kaiser 2021, 2023) by using verb bias effects to diagnose participants’ binding preferences. As mentioned in Section 2.1, if participants prefer non-local binding of *ziji*, self-directed verbs that bias the local referent should elicit reading slowdowns relative to other-directed verbs at/after the onset of the reflexive; likewise, if participants prefer local binding, other-directed verbs should cause reading slowdowns relative to self-directed verbs at/after the reflexive. The biased verbs used in this study are identical to those used by Lyu and Kaiser (2023) where the verb bias manipulation has been found effective in non-topic-biasing contexts.

In our design, the non-local antecedent is always a discourse topic. Controlling the discourse topic status of the non-local antecedent is important, given earlier work which points to the presence of topicality effects. More specifically, when the non-local antecedent is non-topical, *ziji* has been found to prefer local binding (e.g., Dillon et al. 2014, 2016; Jäger et al. 2015; Wang 2017a, b), presumably due to reduced processing costs associated with interpreting a reflexive as a syntactic reflexive over a logophoric reflexive (e.g., Reuland 2001, 2011). This suggests that native Chinese speakers’ default processing strategy is probably to treat *ziji* as a syntactic reflexive, unless discourse or pragmatic constraints intervene. Indeed, when the non-local antecedent is a topic (perspective center), *ziji* has been found to prefer non-local binding (Lyu and Kaiser 2021, 2023). Thus, we expect the *verb bias* manipulation in Experiment 1b to also reveal that *ziji* has some level of preference for the topical, non-local antecedent. In particular, slowdowns with self-directed verbs relative to other-directed verbs at/after the reflexive region would indicate a preference for the topical, non-local antecedent. Absence of a verb bias effect would suggest that the topical, non-local antecedent is *not* favored over the non-topical, local antecedent. (The antecedent choice patterns in Experiment 1a do not speak to the issue of topicality because, unlike the online Experiment 1b, it does not have a diagnostic ‘tool’ to examine this question.)

In the 24 sets of target stimuli, the source and perceiver conditions were created by using the matrix predicates ‘tell’ and ‘hear.’ We chose these two verbs because they were used in prior studies (e.g., Kaiser and Runner 2008; Kaiser et al. 2009) and also because ‘tell’ expresses a volitional speech action to a goal. Half of the topical referents were male, and half were female. The local and non-local antecedents have the same stereotypical gender. Topical referents in critical sentences were pronouns to avoid the repeated name penalty (e.g., Gordon et al. 1993; Almor 1999). The 24 target items were distributed into 4 lists using Latin square and mixed with 24 fillers before being pseudorandomly presented to the participants.

In the offline Experiment 1a, all comprehension questions in target trials probe the interpretation of *ziji*. In the self-paced reading Experiment 1b, 12 out of 20 target trials probe the antecedent of the reflexive and 8 probe the comprehension of the context. In Experiment 1b, we will not analyze the post-reading antecedent choices (which show similar patterns as Experiment 1a) but the results can be found in supplementary materials.

5.2 Experiment 1a: forced choice judgment task

5.2.1 Participants

Forty-two native Mandarin Chinese speakers in mainland China above age 18 participated over the internet.

5.2.2 Procedure

The experiment was conducted on the online experiment platform PCIBex (Zehr and Schwarz 2018). On each trial, participants first saw the context sentence. When participants pressed the spacebar, the context sentence disappeared and was replaced by the critical sentence. We presented the context sentence separately from the critical sentence to make the presentation style as similar as possible between the forced-choice and the self-paced reading experiments. After participants had finished reading the critical sentence, a two-alternative comprehension question appeared on the next screen. This question probed the antecedent of the reflexive (e.g., ‘Who does the academic paper belong to?’). Participants then clicked on one of the two referents (e.g., ‘Xiaoming’/‘Prof. Wang’). The order of the two choices was randomized.

5.2.3 Predictions

We predict a main effect of VERB BIAS (e.g., Lu 2011; Liu 2020; Lyu and Kaiser 2021, 2023): participants should prefer a local reading of *ziji* with self-directed verbs and a non-local reading with other-directed verbs. Crucially, a main effect of LOGOPHORIC ROLE is predicted if the Logophoric Hierarchy modulates the binding preferences of *ziji*: participants should prefer a topical referent that is a source over a

topical referent that is a perceiver. We make no predictions regarding an interaction of VERB BIAS $\times$ LOGOPHORIC ROLE.

5.2.4 Data analysis

It was decided ahead of time that participants with comprehension accuracies lower than 75% on filler trials would not be included in data analysis. All participants passed the screening. Mixed effects logistic models were run in R (R Core Team 2022) over participants' binary antecedent choices (local vs. non-local) using the *glmer* function implemented by the R package *lme4*. The factors VERB BIAS and LOGOPHORIC ROLE were sum coded (self-directed: 0.5, other-directed: -0.5 ; source: 0.5, perceiver: -0.5). The fixed effects are the two independent variables. The random effects are random slopes and intercepts grouped by VERB BIAS and LOGOPHORIC ROLE. Statistical models were fitted parsimoniously such that results from a simpler model were reported over a more complex model if the two models do not differ significantly in model comparisons (Bates et al. 2015) for which an alpha value of 0.2 was adopted (Matuschek et al. 2017).

5.2.5 Results

The participants' antecedent choices are shown in Fig. 1. The y-axis represents the proportions of non-local choices. The left panel (boxplot) displays the distribution of participants' non-local choices across conditions; the right panel (bar plot) displays the mean proportions of non-local choices. See Table 1 for statistics.

The antecedent choice results are largely consistent with our predictions (Section 5.2.3). First, Chinese speakers overall prefer non-local binding with other-directed verbs and local binding with self-directed verbs. The main effect of VERB BIAS ($p < 0.001$) suggests that participants are more likely to choose a non-local subject when the embedded verb is other-directed. Second, source subjects are marginally more likely to be selected as antecedents compared to perceiver subjects, but this source preference is only present with self-directed verbs. Indeed, the statistical model shows a VERB BIAS $\times$ LOGOPHORIC ROLE interaction ($p < 0.04$), driven by the fact that the source effect is only significant in the self-directed verb conditions ($\beta = -0.78$, $SE = 0.25$, $t = -3.09$, $p < 0.002$) but not in the other-directed verb conditions ($\beta = 0.037$, $SE = 0.35$, $t = 0.11$, $p = 0.92$). The absence of the source advantage with other-directed verbs could be due to ceiling effects (among other potential reasons).

Table 1 Summary of statistics for Experiment 1a

	Estimate	SE	<i>t</i> value	<i>p</i> value
Verb bias	3.45	0.30	11.41	$< 0.001^*$
Logophoric role	-0.36	0.21	-1.72	0.09
Verb bias $\times$ Logophoric role	0.87	0.42	2.08	$< 0.04^*$

* < 0.05

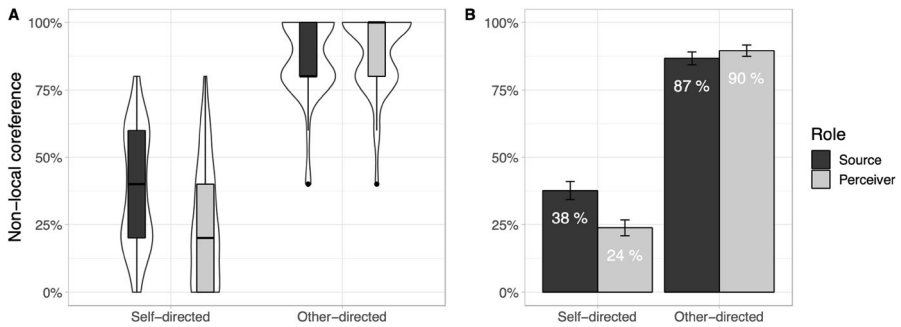


Fig. 1 Interpretation of *ziji* in Experiment 1a. Panel A shows the distribution of all participants' non-local choices by condition. Panel B shows the average proportions of non-local choices by condition for the same data. Error bars show ± 1 SE

Based on the antecedent choice results, we argue that the exempt use of *ziji* is gradiently sensitive to the Logophoric Hierarchy: the gradience is reflected in the fact that the Logophoric Hierarchy does not categorically rule in or rule out binding by the perspective center. This aligns with prior findings for exempt reflexives in Dutch, German, and English (e.g., Kaiser and Runner 2008; Kaiser et al. 2009). Furthermore, our finding that sources are preferred over perceivers shows that the predictions derived from the Logophoric Hierarchy—which Culy (1994) proposed based on logophoric pronouns in pure logophoric languages (e.g., Ewe)—also apply to perspective-sensitive reflexives in mixed logophoric languages (e.g., Chinese). This suggests that both kinds of languages are constrained by the same implicational hierarchy.

However, the offline data do not inform us about *when* different kinds of information are used during real-time processing. We investigate this in Experiment 1b.

5.3 Experiment 1b: self-paced reading

5.3.1 Participants

Eighty-nine native Mandarin Chinese speakers above age 18 participated over the internet. No participant took part in more than one of the studies reported in this paper.

5.3.2 Procedure

The participants did the experiment on PCIBex. Similar to the offline experiment, the self-paced reading experiment shows the context sentence and the critical sentence separately on different screens. Each sentence was displayed chunk by chunk. The critical region is the reflexive *ziji*, followed by three spillover regions. Unlike Experiment 1a, only 3/5 of comprehension questions on critical trials probed the antecedent choices; 2/5 of the questions presented after critical trials probed the

understanding of the context to ensure that participants also paid attention to the context sentence.

5.3.3 Predictions

Prior to the critical region (the reflexive *ziji*) and unrelated to our purposes, we expect reading times (RTs) to differ at/after the local subject (e.g., ‘Prof. Wang’) between the source and perceiver conditions, because the preceding matrix predicates (‘tell others’ vs. ‘hear’) differ. It is possible that the object ‘others’ incurs extra processing costs due to the representation of an additional referent. The presence of an object also makes the sentence longer, which may also yield different RTs. Such differences are irrelevant for the theoretical claims we are testing.

At the reflexive region, the syntax-first and the multiple-constraints accounts make contrasting predictions. First consider the syntax-first account. If real-time reflexive resolution at the initial processing stages is guided by syntactic information (i.e., local search), given the verb bias diagnostic (see Sects. 2.1 and 2.2), we anticipate processing difficulty—longer RTs—to arise when the verb preceding the reflexive is other-directed which conflicts with a local reading of *ziji*. However, we think this scenario is rather unlikely as Lyu and Kaiser (2023) found that Chinese speakers anticipate non-local readings of *ziji* beginning at the earliest moments.

Now, consider what a multiple-constraints account predicts. (As Exp.1 only tests *ziji*, the difference between the uniform vs. form-specific versions is not relevant.) First, we consider the topicality effect. We expect the participants to show reading slowdowns at *ziji* when the preceding verb is self-directed, regardless of the matrix verb type (i.e., ‘tell’ vs. ‘hear’). This is because, as explained earlier, the matrix subject is a discourse topic and likely to be construed as a perspective center (Kuno’s Topic Empathy Hierarchy). We follow Kuno (1987) (see ex.(6)) by assuming that when a perspective center exists, people tend to empathize with that perspective center (the matrix subject). Therefore, Chinese speakers may prefer to bind the perspective-sensitive *ziji* to this topical perspective center. Applying the verb bias diagnostic, self-directed verbs which conflict with non-local binding preferences of *ziji* should elicit slowdowns (but not other-directed verbs which favor non-local binding).

Next, we consider the logophoric role effect. Multiple-constraints accounts lead us to expect an immediate effect of the subject’s logophoric role: the non-local binding preference (longer RTs induced by self-directed verbs) should be stronger in the source than in the perceiver conditions. In other words, we expect a $\text{VERB BIAS} \times \text{LOGOPHORIC ROLE}$ interaction at the reflexive.

5.3.4 Data analysis

Prior to analysis, it was decided that participants whose comprehension question accuracies fall below 75% would be excluded. As a result, 3 participants were removed from analysis. In addition, RTs below 100 ms, over 3000 ms and above 2.5

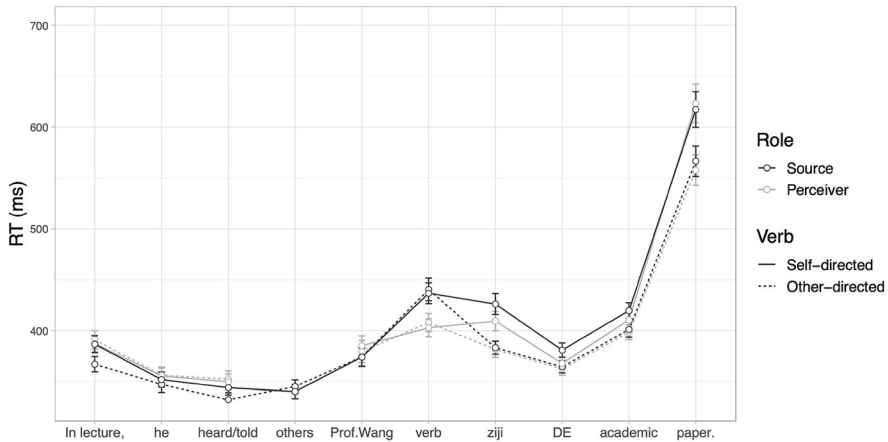


Fig. 2 Raw RTs by region and condition in Experiment 1b

SDs from the mean by region and condition were excluded. This resulted in the removal of 3.26% of the data. Mixed effects linear models were run over log-transformed RTs. Experiments 1a and 1b use the same coding and the same procedure for model selections.

5.3.5 Results

The 86 participants' raw RTs by region and condition are shown in Fig. 2. The summary of statistics is in Table 2. Prior to the critical reflexive region, the biased verb region (following the referent 'Prof. Wang') shows a main effect of LOGOPHORIC ROLE ($p < 0.001$), which persists at the critical region ($p < 0.05$). As predicted and unrelated to our purposes, this may be due to the presence of the matrix object (i.e., 'others') in the source conditions.

At the critical reflexive region, in addition to the effect of LOGOPHORIC ROLE, we find a main effect of VERB BIAS ($p = 0.003$), with self-directed verbs elicited RT slowdowns relative to other-directed verbs. Pairwise comparisons show that this verb bias effect is significant in both source conditions ($\beta = -0.05$, $SE = 0.02$, $t = -2.53$, $p = 0.01$) and perceiver conditions ($\beta = -0.05$, $SE = 0.02$, $t = -2.53$, $p = 0.01$). This suggests that upon encountering *ziji*, the participants preferred non-local binding than local binding regardless of the subject's logophoric role, even when the topical subject is a perceiver and has the least prominent role on the Logophoric Hierarchy.² At the following three spillover regions, the main effect of VERB BIAS persists (p 's < 0.05). There is no significant VERB BIAS $\times$ LOGOPHORIC ROLE interaction

² We note that longer reading times elicited by self-directed verbs are not due to meaning differences between self-directed and other-directed verbs. In fact, previous studies show that when the matrix subject is not a discourse topic, other-directed verbs lead to reading slowdowns, exhibiting a locality bias (Lyu and Kaiser 2023). Additionally, the log frequencies of these two types of verbs do not differ significantly and are not expected to contribute to different processing difficulties.

Table 2 Summary of statistics for log-transformed RTs in Experiment 1b

Region	Verb bias			Logophoric role			Verb bias $\times$ Logophoric role					
	β	SE	t	p	β	SE	t	p	β	SE	t	p
In lecture	-0.02	0.01	-1.80	0.07	0.01	0.01	0.79	0.42	0.05	0.03	1.75	0.08
he	-0.01	0.01	-0.81	0.42	0.02	0.01	1.45	0.15	0.02	0.03	0.78	0.43
told/heard	-0.01	0.01	-1.11	0.27	0.02	0.02	1.33	0.20	0.01	0.03	0.41	0.68
others	0.02	0.02	0.98	0.33	-	-	-	-	-	-	-	-
Prof. Wang	-0.02	0.02	-1.21	0.23	-0.004	0.02	-0.24	0.81	-0.04	0.03	-1.30	0.20
verb	0.003	0.02	0.18	0.86	-0.06	0.02	-3.63	< 0.001*	0.03	0.03	0.96	0.34
ziji	-0.05	0.02	-3.09	0.003*	-0.03	0.01	-1.98	< 0.05*	0.0004	0.03	0.01	0.99
DE	-0.03	0.01	-2.37	0.02*	-0.02	0.01	-1.31	0.19	0.02	0.02	1.02	0.31
academic	-0.04	0.01	-2.49	0.01*	-0.02	0.01	-1.33	0.18	0.02	0.03	0.98	0.33
paper	-0.07	0.02	-3.59	< 0.001*	-0.01	0.02	-0.64	0.53	-0.02	0.04	-0.41	0.68

* < 0.05

at any region of interest. But at the first spillover region (DE), a delayed logophoric role effect emerges: self-directed verbs only caused reading slowdowns in the source conditions ($\beta=-0.04$, $SE=0.02$, $t=-2.32$, $p=0.02$) but not in the perceiver conditions ($\beta=-0.02$, $SE=0.02$, $t=-0.95$, $p=0.35$). This seems to suggest that the logophoric role information is accessed at a slightly later stage relative to discourse topicality.

5.4 Discussion

Using forced choice (Experiment 1a) and self-paced reading (Experiment 1b) experiments, we investigated how two types of discourse-level information—discourse topicality and logophoric roles—impact the resolution of *ziji* in Chinese.

The offline forced choice Experiment 1a replicated prior findings by showing that verb bias plays a major role on antecedent choices (e.g., Lu 2011; Liu 2020; Lyu and Kaiser 2021, 2023). The results show that Chinese speakers (i) strongly favor non-local binding when the biased verb preceding *ziji* is other-directed and (ii) favor local binding when the verb is self-directed. This strengthens the wide-spread view that reflexive resolution is not only driven by syntactic (e.g., locality) factors, but also by semantic factors (see discussion in e.g., Yu 1992; Pollard and Xue 1998). But crucial for our purposes, the verb bias effect is modulated by the logophoric role of the topical subject. We found that the preference for local binding with self-directed verbs is attenuated when the non-local topical subject is a source vs. a perceiver. This suggests that Culy's Logophoric Hierarchy interacts with the binding of *ziji* in a gradient manner. (We attribute the lack of a logophoric role effect in the other-directed verb conditions due to ceiling effects.) As a whole, Experiment 1a provides evidence that *ziji* exhibits similar properties to exempt reflexives in other languages such as Dutch, German, and English.

The online self-paced reading Experiment 1b probes when the two types of discourse-level information—discourse topicality and logophoric role—guide referential processing. Using the verb bias diagnostic, we find that as soon as *ziji* is encountered, Chinese speakers expect *ziji* to be bound by the non-local discourse topic which can be construed as a perspective center given the Topic Empathy Hierarchy (Kuno and Kaburaki 1977; Kuno 1987). This suggests that discourse topicality immediately guides online reflexive resolution (also see Patil et al. 2016 for evidence from English) and that local search is not prioritized (cf. e.g., Sturt 2003; Cunnings and Sturt 2014). Strikingly, the logophoric role information (source/perceiver) does not immediately modulate the processing of *ziji*. Only at the spillover region DE and post-reading judgment (see Experiment 1a and supplementary materials for Experiment 1b) did we discover stronger preferences for source subjects. Given these findings, it seems that not all types of discourse-level information have an equal impact at early processing stages.

6 Experiment 2: logophoricity and the processing of *ta-ziji*

Experiment 1 showed the Logophoric Hierarchy constrains the online and offline interpretation of *ziji* which can function as a perspective-sensitive reflexive. In Experiment 2, we query whether *ta-ziji* exhibits similar properties or whether it differs from *ziji* in terms of its sensitivities to discourse topicality and logophoric roles. Experiment 2a is an offline forced choice task. Experiment 2b is a self-paced reading task.

6.1 Materials and design

The sentence materials and the experiment design are identical to those in Experiment 1, except that *ziji* was replaced by *ta-ziji*.

6.2 Experiment 2a: forced choice judgment

6.2.1 Participants

Forty-four native Chinese speakers over 18 in mainland China participated in this experiment.

6.2.2 Procedure

The same procedure described in Section 5.2.2 was followed in Experiment 2a.

6.2.3 Predictions

We make predictions for the antecedent choice patterns for *ta-ziji* as well as the differences between *ziji* and *ta-ziji*. First, following prior observations (e.g., Yu 1992; Lyu and Kaiser 2023), we anticipate a main effect of VERB BIAS similarly to our predictions for *ziji*: Chinese speakers are expected to prefer local binding with self-directed verbs and non-local binding with other-directed verbs.

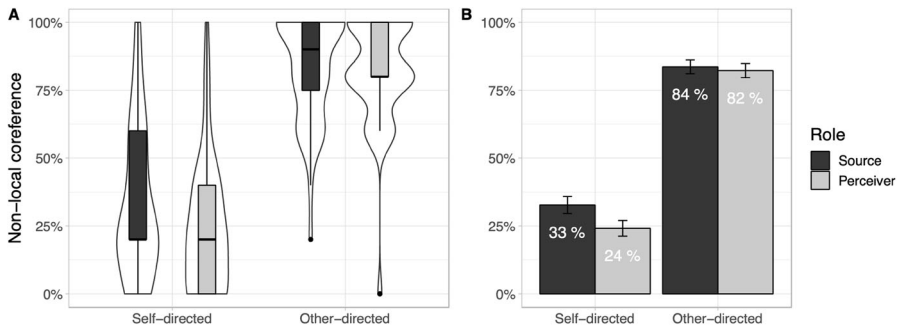
Second, if *ta-ziji* exhibits perspective-sensitivity, it should also be constrained by the Logophoric Hierarchy. Thus, a main effect of LOGOPHORIC ROLE or a VERB BIAS $\times$ LOGOPHORIC ROLE interaction (given Experiment 1a) may emerge. However, if *ta-ziji* is selectively sensitive to certain discourse prominence such as discourse topicality, as shown by Lyu and Kaiser (2023), but not perspective-related logophoric prominence (i.e., source vs. perceiver role), no logophoric role effect is expected either within the self-directed or the other-directed verb conditions.

Finally, when comparing the antecedent choices in Experiment 2a (*ta-ziji*) and Experiment 1a (*ziji*), we may find a stronger preference for non-local binding with *ziji* than with *ta-ziji* if discourse topicality has a stronger impact on *ziji* relative to *ta-ziji*. Additionally, the logophoric role effects shown by *ziji* and *ta-ziji* could differ, although we are agnostic about which reflexive shows a stronger logophoric role effect.

Table 3 Summary of statistics for Experiment 2a

	Estimate	SE	<i>t</i> value	<i>p</i> value
Verb bias	3.37	0.39	8.65	< 0.001*
Logophoric role	−0.34	0.19	−1.75	0.08
Verb bias × Logophoric role	0.42	0.38	1.10	0.27

* < 0.05

**Figure 3** Interpretation of *ta-ziji* in Experiment 2a. Panel A shows the distribution of all participants' non-local choices by condition. Panel B shows the average proportions of non-local choices by condition. Error bars show ± 1 SE

6.2.4 Data analysis

For data analysis on *ta-ziji*, we used the same analysis methods as described in Section 5.2.4. We also conducted analyses comparing the antecedent choices in Experiments 2a and 1a, by including the factor REFLEXIVE TYPE (*ziji*: 0.5, *ta-ziji*: −0.5) in an additional mixed effects logistic model.

6.2.5 Results

The participants' mean preferences for non-local antecedents are displayed in Fig. 3. See Table 3 for statistics. The statistical model shows an expected main effect of VERB BIAS ($p < 0.001$). Non-local binding is preferred with other-directed verbs; local binding is preferred with self-directed verbs. The main effect of LOGOPHORIC ROLE is only marginal ($p = 0.08$). There is no VERB BIAS × LOGOPHORIC ROLE interaction. Nevertheless, planned comparisons reveal an effect of logophoric role with self-directed verbs ($\beta = -0.57$, $SE = 0.25$, $t = -2.30$, $p = 0.02$)—sources attract more non-local choices than perceivers—but not with the other-directed verbs ($\beta = -0.15$, $SE = 0.30$, $t = -0.49$, $p = 0.62$).

Overall, statistical analyses indicate that *ta-ziji* behaves similarly to *ziji* in terms of the verb bias effect and the logophoric role effect. Thus, Experiment 2a provides concrete evidence suggesting that *ta-ziji* can indeed be used as a perspective-

Table 4 Summary of statistics comparing antecedent choices in Experiments 1a and 2a

	Estimate	SE	<i>t</i> value	<i>p</i> value
Verb	3.06	0.15	20.86	< 0.001*
Role	−0.29	0.13	−2.20	0.03*
Reflexive	0.30	0.14	2.16	0.03*
Verb × role	0.60	0.27	2.25	0.02*
Verb × reflexive	0.39	0.27	1.47	0.14
Role × reflexive	−0.05	0.27	−0.18	0.86
Verb × role × reflexive	0.48	0.53	0.90	0.37

* < 0.05

sensitive reflexive constrained by the Logophoric Hierarchy, similar to complex reflexives in Dutch, German, and English.

However, based on visual inspections of Figs. 1 and 3, it appears that, relative to *ziji*, *ta-ziji* shows a weaker discourse topicality effect (i.e., lower proportions of non-local choices) and a slightly weaker source effect within the self-directed verb conditions (9% for *ta-ziji* vs. 14% for *ziji*). To examine the potential differences between these two reflexive forms, we included REFLEXIVE TYPE as a between-group factor.

The summary of statistics from a mixed effects model with factors VERB BIAS, LOGOPHORIC ROLE, and REFLEXIVE TYPE is in Table 4. As expected, the main effect of VERB BIAS ($p < 0.001$) is significant. Additionally, the main effect of LOGOPHORIC ROLE also reaches significance ($p=0.03$). Crucial for our purposes, participants' preference for non-local antecedents is overall stronger with *ziji* than with *ta-ziji*, as indicated by a main effect of REFLEXIVE TYPE ($p=0.03$). This replicates previous findings by Lyu and Kaiser (2023) where *ziji* shows stronger non-local binding preferences than *ta-ziji* in contexts with topical matrix subjects but not in neutral contexts without topical referents. Finally, similar to what we observed in Experiment 1a, the VERB BIAS × LOGOPHORIC ROLE interaction is also significant ($p = 0.02$). No other effects are significant. Given these results, we argue that while *ziji* is more likely to be influenced by discourse topicality, there is no evidence that the two reflexive forms differ in their sensitivity to the logophoric role of the perspective center.

6.3 Experiment 2b: self-paced reading

6.3.1 Participants

Eighty native Mandarin speakers over age 18 participated over the internet.

6.3.2 Procedure

The procedure for Experiment 2b is the same as in Experiment 1b.

6.3.3 Predictions

First, for regions prior to the critical reflexive region, similar to our predictions for Experiment 1b, we expect longer RTs for the source conditions at/after the embedded subject (e.g., ‘Prof. Wang’) due to differences in sentence length and referential complexity (i.e., ‘others’ in source conditions).

Next, we make predictions for how and when discourse topicality and logophoric roles could impact the real-time processing of *ta-ziji*. For the discourse topicality effect, we find it important to highlight the findings of Lyu and Kaiser (2023) on *ta-ziji*, reviewed in Section 2.1, as the predictions for this study are grounded in that work.

Using sentences with a mixture of perception, speech, and belief matrix predicates, Lyu and Kaiser (2023) found that discourse topicality plays an *immediate* role when readers encounter *ta-ziji*. When the matrix subject is *non-topical* (i.e., there is no prior context introducing the subject as a topic), Chinese readers showed a clear locality effect, indicated by longer RTs with other-directed verbs than with self-directed verbs. However, Lyu and Kaiser also found that when the matrix subject is *topical* (i.e., mentioned in the prior sentence, as in Experiment 2b here), there was no effect of verb bias: there were no RT differences for self-directed verbs biasing local subjects vs. other-directed verbs biasing non-local subjects. Intuitively, it seems like being a topic boosts the non-local subject’s accessibility.

Based on these results, if discourse topicality has an early effect in processing, then—at least for the perceiver conditions in Experiment 2b—we anticipate no effect of verb bias or at most a weak locality effect (i.e., other-directed verbs causing slowdowns), given that the non-local subjects in Experiment 2 are topical. In other words, since Lyu and Kaiser (2023) found that conditions with non-topical vs. topical non-local subjects diverged early on, with sentences having topical non-local subjects yielding comparable RTs for self- and other-directed verb conditions, we also interpret a lack of a verb bias effect at the reflexive as a measure of sensitivity to discourse topicality.

As for how logophoric role impacts the processing of *ta-ziji*, there are three possible scenarios. First, it could be that logophoric roles are immediately used such that Chinese speakers prefer non-local binding the moment they encounter *ta-ziji* (i.e., self-directed verbs causing slowdowns) when the topical referent is a source. Together with our prediction for the absence of (strong) verb bias effect in the perceiver conditions, this would lead us to expect a VERB BIAS $\times$ LOGOPHORIC ROLE interaction at the reflexive, which is what the uniform multiple-constraints account predicts. Second, the logophoric role effect could be delayed to the spillover regions, similar to Experiment 1b, which would lead us to expect an interaction at the spillovers—no verb bias effect for perceiver conditions but a non-local binding preference for source conditions. Finally, the logophoric role effect may not appear during online processing, which would mean that logophoric roles do not modulate the online processing of *ta-ziji*. To these predictions, we add that any processing differences between *ziji* and *ta-ziji* in terms of the magnitude and/or timing of discourse effects would be in line with a form-specific approach.

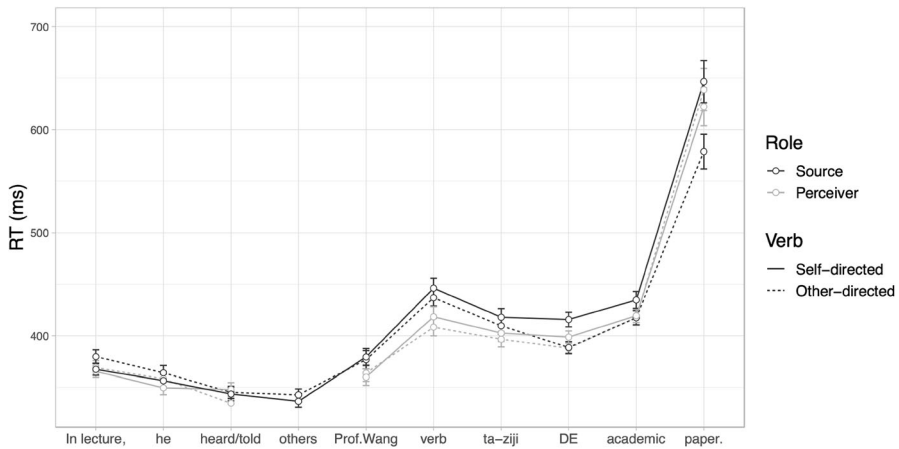


Fig. 4 Raw RTs by region and condition in Experiment 2b

6.3.4 Data analysis

The same data filtering, coding, and analysis methods described in Section 5.3.4 were applied to Experiment 2b. Seven participants did not pass the screening as their accuracy rate on the comprehension questions on filler trials was below 75%, leaving 73 participants for the final analysis. Trimming based on RTs, as done in Experiment 1b, affected 3.33% of the original data.

6.3.5 Results

Figure 4 plots the participants' mean RTs by region and condition. See Table 5 for statistics. Prior to the critical reflexive region, an expected main effect of LOGOPHORIC ROLE reaches significance at the embedded subject (e.g., 'Prof. Wang') and the biased verb ($p < 0.01$), due to an additional referent 'others' in the source conditions.

At the reflexive region, there is no main effect of VERB BIAS. As discussed in Section 6.3.3, we take this as evidence that discourse topicality plays an early role in the processing of *ta-ziji*, because when the non-local subject is non-topical, a locality bias should emerge (see Lyu and Kaiser 2023). In fact, lack of verb bias effects characterizes all spillover regions in the perceiver conditions, which contrasts sharply with prior studies which do not have discourse-topic-biasing contexts and find locality biases instead (e.g., Qian and Wu 2016; Dillon et al. 2016; Wang 2017a, 2017b; Chang et al. 2020). There is no VERB BIAS $\times$ LOGOPHORIC ROLE interaction, aligning with the findings for *ziji* in Experiment 1b.

At the first spillover region (DE), a main effect of VERB BIAS was found ($p < 0.001$); self-directed verbs elicited reading slowdowns. However, although there is no significant VERB BIAS $\times$ LOGOPHORIC ROLE interaction, planned comparisons show that it is only in the source conditions that self-directed verbs caused RT slowdowns

Table 5 Summary of statistics for log-transformed RTs in Experiment 2b

Region	Verb bias				Logophoric role				Verb bias $\times$ Logophoric role			
	β	SE	t	p	β	SE	t	p	β	SE	t	p
In lecture	0.01	0.01	1.15	0.25	-0.02	0.01	-1.46	0.14	-0.02	0.03	-0.92	0.36
he	0.01	0.01	1.06	0.29	-0.02	0.01	-1.76	0.08	0.005	0.03	0.17	0.87
told/heard	-0.009	0.01	-0.68	0.50	-0.005	0.01	-0.41	0.68	-0.04	0.03	-1.70	0.09
others	0.02	0.02	1.27	0.21	-	-	-	-	-	-	-	-
Prof. Wang	0.002	0.02	0.11	0.91	-0.04	0.02	-2.53	0.01*	0.02	0.03	0.53	0.60
verb	-0.01	0.02	-0.60	0.55	-0.07	0.02	-4.52	< 0.001*	0.002	0.03	0.06	0.95
<i>ta-ziji</i>	-0.02	0.01	-1.15	0.25	-0.03	0.02	-1.78	0.08	0.01	0.03	0.47	0.64
DE	-0.04	0.01	-3.31	< 0.001*	-0.02	0.01	-1.36	0.18	0.04	0.02	1.46	0.15
academic	-0.02	0.01	-1.72	0.09	-0.02	0.01	-1.24	0.22	0.02	0.03	0.73	0.47
paper	-0.04	0.02	-1.85	0.06	0.03	0.02	1.53	0.13	0.09	0.04	2.12	0.03*

* < 0.05

($\beta = -0.06$, $SE = 0.02$, $t = -3.29$, $p = 0.001$). In the perceiver conditions, the verb bias effect is not significant ($\beta = -0.02$, $SE = 0.02$, $t = -1.38$, $p = 0.17$).

Finally, at the wrap-up region, only the *VERB BIAS* $\times$ *LOGOPHORIC ROLE* interaction reaches significance ($p < 0.03$): we again find that the verb bias effect (i.e., non-local binding preference) is only significant in the source conditions ($\beta = -0.08$, $SE = 0.03$, $t = -2.82$, $p = 0.005$) but not in the perceiver conditions ($\beta = 0.01$, $SE = 0.03$, $t = 0.33$, $p = 0.74$). Thus, together with similar results from Experiment 1b, the logophoric role effect seems delayed relative to the discourse topicality effect. But crucially, Experiment 2b provides additional novel evidence that *ta-ziji* tends to be used as a non-local exempt reflexive when its antecedent is a topical source.

6.4 Discussion

The goals of Experiment 2a and 2b are to examine *ta-ziji*'s binding preferences in the presence of topically and logophorically prominent (i.e., source vs. perceiver) antecedents, as well as to shed light on the relative timing of the accessibility of discourse topicality and logophoric role information.

Experiment 2a shows that the antecedent choice patterns elicited by *ta-ziji* resemble those of *ziji* in Experiment 1a. In addition to the expected verb bias effect (also found by Lyu and Kaiser 2023), we observe stronger preferences for source topical subjects within the self-directed verb conditions, conforming to the Logophoric Hierarchy. This suggests that, like complex reflexives in other languages (Kaiser and Runner 2008; Kaiser et al. 2009; Sloggett 2017), *ta-ziji* is also constrained by the Logophoric Hierarchy originally proposed for pure logophoric languages. However, Experiment 2a only reveals whether *ta-ziji* is sensitive to Logophoric Hierarchy but does not inform us about the binding preferences—whether *ta-ziji* tends to be used as an exempt reflexive referring to perspective centers—in different discourse contexts.

To investigate the binding preferences of *ta-ziji* during incremental processing, we used the verb bias effect as a diagnostic tool: reading slowdowns elicited by self-directed and other-directed verbs indicate non-local and local binding preferences, respectively. We find that, within the perceiver conditions, when the matrix subject is topically prominent but not prominent on the Logophoric Hierarchy, Chinese speakers do not show a preference for either local or non-local binding. This largely replicates Lyu and Kaiser's (2023) findings but, crucially, does not suffer from the complication of their design which included matrix subjects with different logophoric roles (e.g., perceiver, thinker, source). Thus, based on their study one cannot draw conclusions about effects of logophoric roles.

Interestingly, this weaker discourse topicality effect shown by *ta-ziji* (i.e., absence of verb bias effect) relative to *ziji* in online processing echoes our comparative analysis on Chinese speakers' antecedent choices in Experiments 1a and 2a, where we found weaker preferences for non-local binding with *ta-ziji* compared to *ziji*. This is compatible with a form-specific multiple-constraints view on offline reflexive resolution (see Section 7 for more discussions on form-related differences).

Critically, we find evidence from participants' RTs that when the topical matrix subject has the logophoric role of source, *ta-ziji* tends to be used as an exempt reflexive.³ To our knowledge, few experimental studies have explored whether and how discourse factors can give rise to the exempt use of *ta-ziji* bound by a perspective center. The RTs also shed light on when the two types of discourse-level information guide real-time processing. Similar to what we found in Experiment 1b on *ziji*, the processing pattern of *ta-ziji* reveals a delayed logophoric role effect, relative to the immediate discourse topicality effect (when Lyu and Kaiser's 2023 study is taken into account, as discussed in Sects. 3 and 6.3.3). Given these findings, we propose that different subtypes of discourse-level information may be accessed at different time points during reflexive resolution, just like different classes (e.g., syntactic vs. semantic/discourse) of information may be accessed at different time points (e.g., Dillon et al. 2013; Cunnings and Sturt 2014). These results present challenges for the uniform multiple-constraints account.

Taken together, Experiments 2a and 2b yield clear evidence that *ta-ziji* can be used as a logophoric reflexive when the non-local antecedent is a sufficiently prominent perspective center.

An anonymous reviewer asks whether *ta-ziji* could be interpreted by participants as an intensified reflexive. This is an important question, as intensification could be (partially) responsible for its referential properties. However, we think an intensifier interpretation is unlikely in our experiments for three main reasons. First, due to lack of any contextual support: As discussed extensively by Pan (1998), for *ta-ziji* to be interpreted as an intensifier, it has a contrastive reading and requires contextual support (see examples in Pan 1998, 791). But in our experiments, the context does not support a contrastive reading, rendering an intensified reading unlikely. Second, due to its syntactic position: As noted by Xu (1999) (also see Wang (2017a)), *ta-ziji* when used as an intensifier typically occurs in subject position. But in our studies, *ta-ziji* was in genitive position modifying a direct object, thus rendering an intensified reading even more unlikely. Third, due to processing load: It is widely-agreed that more marked interpretations typically entail greater processing load. Thus, interpreting *ta-ziji* as an intensifier, a marked and infrequent usage, would entail more processing effort, contrary to the principle of processing economy (e.g., Bever 1970; Frazier and Fodor 1978; Frazier 1979). Thus, given these reasons, we think that participants are unlikely to have assigned an intensified reading to *ta-ziji* in our experiments.

³ Experimental work suggests that, for both *ziji* and *ta-ziji*, when the non-local referent is not discourse-prominent, the reflexive tends to refer to the local antecedent, which means that the reflexives, by default, tend to be used as local anaphors (e.g., Jäger et al. 2015; Dillon et al. 2016; Lyu and Kaiser 2023), unless discourse factors are introduced (Pollard and Xue 2001).

7 General discussion

7.1 Summary of findings

The experiments reported in this paper aim to contribute to our understanding of the logophoric properties of the two Chinese reflexives, *ziji* and *ta-ziji*, as well as to explore how different types of discourse-level information are recruited during online reflexive resolution. We conducted two sets of offline forced choice and online self-paced reading experiments. Before summarizing our findings, we first consider an important issue raised by an anonymous reviewer concerning the interpretation of gradient data and its relationship to linguistic theories.

In this study, we examined whether *ziji* and *ta-ziji* can function as exempt reflexives by assessing native speakers' binding preferences during real-time processing. However, a question arises: how can we shed light on potentially categorical grammatical constraints based on preferences? In experimental syntax, it is widely accepted that behavioral data from study participants often exhibit gradient patterns even if the underlying linguistic mechanism is categorical (e.g., McDaniel and Cowart 1999; Alexopoulou and Keller 2007; Sprouse et al. 2012, 2016). This is due to various factors, including experimental noise, processing factors, and individual variation (see Schütze and Sprouse 2014 for more discussion). In light of this, experimental studies often rely on gradient measures (e.g., forced choice, acceptability judgment, magnitude estimation) to gain insights into categorical grammatical constraints (e.g., Cunnings and Sturt 2014; Kush et al. 2015; Villata et al. 2016; Bruening and Al Khalaf 2019; Hammerly 2022; Salzmann et al. 2023). Crucially, we note that the gradient nature of our data does not undermine the theoretical contribution of this study. With this clarification, we now summarize our findings before discussing broader issues.

First, let us consider the findings of Experiments 1a and 2a which examined the offline interpretation of *ziji* and *ta-ziji*, respectively. In these offline forced choice tasks, in addition to an expected, strong verb bias effect—local binding is preferred with self-directed verbs and non-local binding is preferred with other-directed verbs—we found stronger preferences for topical source referents over topical perceiver referents, at least within the self-directed verb conditions. This crucially suggests that both forms of reflexives are constrained by the Logophoric Hierarchy and highlights the similarities between logophoric pronouns in pure logophoric languages and long-distance exempt reflexives in mixed logophoric languages such as Chinese. The logophoric role effect for *ta-ziji* is especially important: unlike previous studies which either found only indirect links between perspective-taking and the exempt use of *ta-ziji* or are subject to alternative explanations (Liu 2020; Lyu and Kaiser 2023), Experiment 2a yields strong evidence that *ta-ziji* can be used as an exempt reflexive constrained by the Logophoric Hierarchy.

However, *ziji* and *ta-ziji* differ slightly in terms of the antecedent choice patterns: Chinese speakers overall chose non-local subjects significantly more often with *ziji* than with *ta-ziji*. Given the link between discourse topicality and perspective-taking (e.g., Kuno 1987), *ziji* may be more easily used as a perspective-sensitive exempt

reflexive than *ta-ziji*. Viewed from this perspective, the difference between *ziji* and *ta-ziji* may stem from long-distance *ta-ziji* needing a stronger or more prominent perspective center. Nevertheless, Experiments 1a and 2a show that these two reflexives are more similar than previously thought; the interpretation of both is sensitive to the logophoric source/perceiver status of the antecedent.

Next, we turn to the findings of the self-paced reading Experiments 1b and 2b on the processing of *ziji* and *ta-ziji*, respectively. For these two experiments, we followed previous studies (e.g., Lyu and Kaiser 2021, 2023) by using verb bias effects as a diagnostic tool to test Chinese speakers' binding preferences when they encounter the reflexive. Specifically, if self-directed verbs cause reading slowdowns relative to other-directed verbs, we take this as evidence that non-local binding is preferred as only self-directed verbs (but not other-directed verbs) conflict with non-local binding. Likewise, if other-directed verbs cause slower reading times, this indicates a preference for local binding.

On the processing of *ziji*, we find that regardless of the logophoric prominence (source vs. perceiver) of the topical subject, Chinese speakers treat *ziji* as an exempt reflexive referring to a non-local antecedent the moment *ziji* is encountered, indicated by reading slowdowns with self-directed verbs relative to other-directed verbs. This aligns with the findings of prior studies which suggest that discourse topicality information is immediately accessible to the parser in reflexive resolution (e.g., Patil et al. 2016; Lyu and Kaiser 2021, 2023). In addition, mirroring the offline Experiment 1a where a logophoric role effect was identified in antecedent choices, the RTs also show that Chinese speakers have an even stronger preference for non-local binding when the topical subject is a source, compared to when it is a perceiver. However, the logophoric role effect seems to modulate the processing of *ziji* at a somewhat later stage (i.e., first spillover region), which suggests that the two types of discourse-level information are utilized at different stages.

The processing profile of *ta-ziji* resembles *ziji* in two aspects. First, upon encountering the reflexive, the participants were sensitive to the topical status of the matrix subject; they did not show a preference for the local or non-local subject—i. e., no differences between self- vs. other-directed verbs. We take this as evidence that discourse topicality plays an early role in the processing of *ta-ziji* by boosting the prominence of the topical, non-local subject. More specifically, we interpret the lack of a locality preference against the backdrop of other work showing that when the non-local subject is *not topical*, there *is* an early locality bias effect (other-directed verbs elicit slowdowns) at the reflexive region (Lyu and Kaiser 2023).⁴ Our results suggest that local search does not always constrain reflexive resolution (also see Principle Z by Pollard and Xue 2001) and that discourse-level information can

⁴ As Experiment 2b did not test 'neutral context' conditions, we cannot exclude the possibility that the absence of a verb bias effect could be due to factors unrelated to topicality (e.g., delayed key presses). However, as mentioned in Section 2.2, the stimuli used in Experiment 2b closely match those used in Lyu and Kaiser (2023). The only differences are that in Experiment 2b, (i) all critical sentences were preceded by topical sentences and (ii) the matrix predicates were carefully controlled and classified as source vs. perceiver predicates. Thus, together with the early biased verb effect for *ziji* in Experiment 1b, we think that the absence of a verb bias effect in the perceiver conditions for *ta-ziji* in Experiment 2b can be taken as indicating early effects of the matrix subject's discourse topic status.

be utilized at an early stage, which fits well with multiple-constraints approaches to reflexive processing. Second, we find that with *ta-ziji*, the logophoric role effect is delayed to the first spillover region where a non-local binding preference emerges for the source conditions. This suggests that effects of different subtypes of discourse-level information can ‘kick in’ at different time points.

However, the reading time results for *ziji* and *ta-ziji* differ in one aspect, which echoes what we found for their antecedent choices. At the reflexive region, when the topical subject is a logophorically non-prominent perceiver, *ziji* exhibits a non-local binding preference while *ta-ziji* does not show any preference. Together with the lower proportions of non-local choices for *ta-ziji* compared to *ta-ziji* in the offline experiments (see Section 6.2.5), this suggests that an exempt reading is more available with *ziji* than *ta-ziji*. Furthermore, the different discourse topicality effects for the processing of *ziji* and *ta-ziji* support a form-specific approach to online reflexive resolution.

To summarize, the experiments reported in this paper further our knowledge about the linguistic properties of the two Chinese reflexives and their online processing profiles. On the linguistic front, we have obtained more direct evidence that both *ziji* and *ta-ziji* can be used as perspective-sensitive exempt reflexives when the non-local subject is sufficiently prominent at the discourse level. On the psycholinguistic front, we have shown that discourse topicality can impact the processing patterns of *ziji* and *ta-ziji* differently, which supports a form-specific view of the multiple-constraints account. Additionally, we find that different types of discourse information (i.e., discourse topicality vs. logophoric role) can impact the processing of these two reflexives differently. This points to the possibility that different referential forms can differ in how sensitive they are to different subtypes of discourse-level information. Thus, we provide evidence for form-specific effects on a more fine-grained level than those observed in prior work: not only can different referential forms differ in how sensitive they are to syntactic vs. perspectival/discourse-level information (e.g., Kaiser et al. 2009); they can also exhibit differential sensitivities to subtypes of discourse-level information (e.g., topicality).

7.2 Broader implications for linguistic theories on long-distance reflexives

In this subsection, we consider how our findings bear on linguistic theories of long-distance reflexives, especially in the context of Chinese reflexives. Here, we focus on two accounts, the LF movement account for long-distance *ziji* and Pan’s (1998) prominence-based account for long-distance *ta-ziji*. We argue that these two accounts cannot (fully) predict the outcomes of this study and that a logophoricity-oriented approach to long-distance binding has more merits.

In the theoretical literature, different approaches have been proposed to account for long-distance binding of *ziji*. Among them, the LF movement approach has been particularly influential (e.g., Battistella 1989; Cole et al. 1990; Huang and Tang 1991; Cole and Sung 1994; Cole and Wang 1996; Huang and Liu 2001). The LF movement accounts link the long-distance property of *ziji* to (the lack of) phi-features. As *ziji* is unspecified for person and number, under one version, it can

undergo head movement to adjoin to the TP-head in the local domain and inherit the phi-features of the local subject. From there, it can optionally move to a higher domain and co-index with a higher c-commanding subject (e.g., Cole et al. 1990), which derives long-distance binding. Under another version, *ziji* is a full XP and cyclically moves to adjoin to the local TP and is co-indexed with the subject of a higher clause (e.g., Huang and Tang 1991). Note that the LF movement approach does not have to be a ‘pure’ syntactic theory. In fact, Huang and Liu (2001) posit that Sells’ logophoric role (i.e., source, self, pivot) can be syntactically represented, as shown in ex.(12). However, the LF movement accounts assume that only bare reflexives can be long-distance bound because only bare reflexives lack phi-features and/or are heads. But as our study has shown, the complex reflexive *ta-ziji* (which has a person feature) not only can but also tends to be used as a long-distance reflexive when the matrix subject is a prominent perspective center, which is unpredicted by LF movement accounts.

- (12) Zhangsan₁ shuo [_{SourceP} ziji₁ [Lisi piping-le t₁]].
 Zhangsan say self Lisi criticize-ASP
 ‘Zhangsan₁ said Lisi criticized self₁.’

How about Pan’s (1998) Anaphor Condition which allows long-distance binding of *ta-ziji*? It seems that Pan’s approach does not fully capture the data presented in our study either. The Anaphor Condition essentially states that non-local binding is allowed only when the non-local antecedent is semantically more prominent than the local antecedent, where semantic prominence is determined by Chou’s (1992) Animacy Hierarchy ([+human] > [−human, +animate] > [−animate]). However, the Anaphor Condition is silent about discourse prominence which includes discourse topic prominence and logophoric prominence. Consequently, it does not predict the long-distance binding preference of *ta-ziji* in Experiment 2b. One simple modification would be to incorporate discourse prominence hierarchies (e.g., [+topic] > [−topic], [+source] > [−source], etc.) into Pan’s Anaphor Condition. But doing so misses a crucial point: as suggested in Section 1, Pan’s examples can be explained under a logophoricity account without including a cluster of independent prominence hierarchies into the Anaphor Condition.

Recall ex.(2), where non-local *ta-ziji* is licensed when the local subject is an inanimate DP ‘book.’ As alluded to in Section 1, the Humanness Empathy Hierarchy (Kuno and Kaburaki 1977; Kuno 1987) can readily account for the exempt reading of *ta-ziji*, because (i) people tend to take the perspective of a human rather than that of an inanimate entity and (ii) *ta-ziji* can be used as an exempt reflexive if the non-local subject is a stronger perspective center than the local subject. Similarly, the logophoricity account can also explain the stronger non-local binding preference in the source conditions relative to the perceiver conditions in our study exactly because a topical source is a stronger perspective center than a topical perceiver. Furthermore, Yu’s (1992, 289) observation that *ta-ziji* can be

sentence-free further highlights the relevance of logophoricity for long-distance binding of *ta-ziji*, reproduced in ex.(13).⁵

- (13) Ta-ziji shenghuo shifen jianku, jingchang ba qian yong zai
 He-self live very hard often BA money use at
 bangzhu zhouwei de ren.
 help around DE people
 ‘Self led a very difficult life and often used money to help people
 around him.’

Given the above discussion and the experimental findings of this study, we argue that *ta-ziji*, just like *ziji* and exempt reflexives in other languages (Dutch, German, English, etc.), can be used as an exempt reflexive when a (sufficiently strong) perspective center is present. As such, this study has clear implications for linguistic theories of logophoricity and long-distance anaphora. Indeed, our findings regarding the logophoric properties of *ziji* and *ta-ziji* are compatible with more recent theories on syntactic representations of perspective (e.g., Nishigauchi 2014; Charnavel and Sportiche 2016; Charnavel 2019, 2020). These theories propose a covert logophoric pronoun pro_{log} (or $pro_{axis}/pro_{sentient}$) which sits at the left periphery of the local domain and locally binds the reflexive. The logophoric nature of long-distance binding is through the mediation of the pro_{log} which searches for a perspective center. We think this is a promising approach to analyzing long-distance anaphora in Chinese.

7.3 A unified view on the different approaches to reflexive resolution

So far, we have discussed the implications of our results for linguistic theories of long-distance anaphora. However, anaphora resolution is a dynamic process where the language parser recruits different types of information on the fly to locate or retrieve an antecedent. Thus, it is important to understand how linguistic information is mentally represented and accessed during online processing. In this subsection, we explore the possibility of unifying different strands of psycholinguistic approaches to anaphora resolution.

We have considered three accounts on anaphora resolution, the syntax-first account and two types of multiple-constraints accounts (uniform, form-specific). These approaches revolve around the key debate of whether syntactic constraints are accessed first to the exclusion of other non-syntactic constraints, including semantic

⁵ *Ta-ziji* in this example bears a non-contrastive prosody. In fact, *ta-ziji* with a contrastive prosody makes the sentence odd as there is no other character to contrast with.

and discourse-pragmatic constraints.⁶ Viewed from this angle, it seems that the syntax-first account and the uniform multiple-constraints account represent two opposite ends of the argument: the former asserts the dominant influence of syntactic constraints at early-stage processing, and the latter allows for multiple kinds of information to be used immediately.

As we have seen, the self-paced reading studies we conducted point to a more nuanced situation. On the one hand, the local subject is not the preferred antecedent during online processing. On the other hand, we also see hints that effects of antecedents' logophoric role kick in slightly later than the discourse topicality effect. In fact, a key finding of this study is that online anaphora resolution is guided not only by the linguistic properties of different reflexives within a language, supporting a form-specific account, but also by the types of (discourse-level) information in question, aligning with a constraint-based approach to anaphora resolution (e.g., Kaiser et al. 2009; Dillon et al. 2013; Kush et al. 2017) that allows for different constraints to impact processing differently, except that the constraints we examined belong to the same module.

In sum, an architecture that assumes that (i) multiple constraints guide anaphora resolution and that (ii) it is possible for different anaphoric forms to show different sensitivities to different constraints can best account for both the findings of the present study and crosslinguistic differences of anaphora resolution in prior work. For example, unlike English reflexives which display strong 'intervention' effects (e.g., Pollard and Sag 1992; Pollard and Xue 2001; Sturt 2003; Cunnings and Sturt 2014) where long-distance binding across animate antecedents is not allowed, Chinese reflexives, including *ta-ziji*, show a much weaker intervention effect (e.g., Yu 1992, 288). Thus, it seems natural to speculate that online anaphora resolution may show different processing patterns depending on the types of reflexives, constraints, and languages involved. In some languages (e.g., English), reflexives seem to be relatively more sensitive to syntactic constraints, but reflexives in other languages can be relatively more sensitive to discourse-level information. That said, we note that one challenge for the form-specific and constraint-based approaches to online reflexive resolution lies in working out the weights of specific constraints at play before testable, computationally implementable predictions can be made (e.g., Dillon et al. 2013; Jäger et al. 2015; Parker 2019), which requires a deeper understanding of the linguistic properties of pronominal elements. We regard this as an important next step for the psycholinguistic research on anaphora resolution.

⁶ An anonymous reviewer requests clarification on the relationship between linguistic and psycholinguistic constraints. We consider psycholinguistic constraints to be algorithms that utilize linguistic representations/cues (e.g., c-command, subjecthood, agreement features, semantic gender) to determine what kind of linguistic information should be accessed first during real-time reflexive resolution. The three approaches to reflexive resolution reviewed in this paper are concerned with this inquiry. However, for the purposes of this study, we do not focus on the intricate details of (computationally) implementing reflexive resolution within psycholinguistic models, such as the cue-based retrieval model (e.g., Lewis and Vasishth 2005; Jäger et al. 2015). For details on how reflexive resolution in discourse might be operationalized under a cue-based retrieval framework, see Lyu and Kaiser (2021) for discussion.

8 Conclusion

We report a series of experiments using forced choice judgment and self-paced reading to explore the logophoric properties of the two reflexives in Chinese, *ziji* and *ta-ziji*, as well as their real-time processing profiles. Both methods provide clear evidence that long-distance binding of *ziji* and *ta-ziji* is subject to the influence of the Logophoric Hierarchy, which sheds light on the relation between logophoricity and long-distance anaphora in Chinese. Furthermore, Chinese speakers' reading times show that different subtypes of discourse prominence can modulate online reflexive resolution at different time points and that the same type of discourse information (discourse topicality) can impact the processing of different reflexives differently.

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Author contributions Jun Lyu: Conceptualization, methodology, data curation, investigation, formal analysis, validation, writing—original draft, review and editing. Elsi Kaiser: Conceptualization, supervision, validation, writing—original draft, review and editing.

Data availability The data, analyses, and supplementary materials are available at https://osf.io/647ur/?view_only=726a042068e746699dc871b250654b6c.

Declarations

Competing interests The authors declare that they have no competing interests.

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